

Dataset	Size	SOTA	MDE Binomial	MDE (Lachenbruch, 1992)	MDE regression	$\hat{\Delta}$	$ \Delta $ (std.err., n)	Δ_{SOTA}
WNLI	147	94.5%	+5.38%	+5.42% (5.36%, 5.45%)	+5.26%	-1.17%	1.72 (0.917, 4)	0.0%
MRPC	1725	92.0%	+2.40%	+1.91% (0.45%, 2.48%)	+1.62%	+0.03%	+0.625 (0.234, 8)	+0.6%
SST-2	1821	97.2%	+1.34%	+1.10% (0.43%, 1.35%)	+1.02%	+0.18%	+0.571 (0.197, 7)	-0.3%
RTE	3000	91.7%	+1.89%	+1.48% (0.26%, 1.96%)	+1.23%	+1.11%	+3.89 (1.23, 10)	+0.8%
QNLI	5463	97.5%	+0.77%	+0.60% (0.14%, 0.78%)	+0.55%	+0.69%	+1.31 (0.552, 9)	+0.9%
MNLI-m	9796	91.6%	+1.08%	+0.82% (0.08%, 1.12%)	+0.67%	+0.12%	+0.97 (0.442, 10)	+0.2%
MNLI-mm	9847	91.3%	+1.09%	+0.84% (0.08%, 1.14%)	+0.68%	+0.34%	+1.29 (0.550, 8)	+0.3%
QQP	390965	91.0%	+0.18%	+0.13% ($8.45 \times 10^{-5}\%$, 0.19%)	+0.11%	+0.08%	0.36 (0.121, 5)	+0.1%
SQuAD 2.0	8862	90.724%	+1.18%	+0.91% (0.09%, 1.23%)	+0.556%	+0.528%	+2.23% (0.431, 14) †	+0.146%

Table 9: The minimum detectable effect (MDE) for various datasets given the current top accuracy on the leaderboard on May 6th, 2020. See Appendix E for expanded details. How to use this table? Suppose you are building a model to get SOTA on any of these datasets. If you don’t have a reasonable expectation that your model will exceed the MDE, then it is not worth proceeding with the study on a dataset of this size and instead either more data should be collected or a different (larger) dataset used. MDE (Lachenbruch, 1992) provides a mid-point and upper/lower bound assumptions using the most conservative and generous estimates of model agreement. MDE Binomial uses the binomial test as the assumed statistical test and calculates the MDE using the exact mechanism from Appendix E.5. See also discussion by Arend and Schäfer (2019). $\hat{\Delta}$ is the expected effect by fitting a regression to all SOTA improvement claims found in reviewed papers. $|\Delta|$ (std.err., n) is the average improvement in surveyed papers that claimed SOTA and had a positive effect size reported for the dataset (with standard error and the number of papers in parentheses). † indicates that the SQuAD 2.0 average improvement was based on improvements to the SQuAD leaderboard, but weren’t necessarily reported as improvements in a publication. Δ_{SOTA} is the gap between the SOTA model (ALBERT + DAAF + NAS) on GLUE and the next best model (ERNIE) – this was not included in the regression.

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Power	57	0.698	0.352	0.034	0.407	1.000	1.000
P	57	0.220	0.283	0.000	0.00000	0.348	1.000
Statistic	N	Percentage	-				
% Powered	57	0.456%	—	—	—	—	—
% Significant	57	0.509%	—	—	—	—	—
% significant and Powered	57	0.368%	—	—	—	—	—

Table 10: We examine the claims of SOTA improvement in surveyed GLUE papers and use a leave-one-out regression-based estimate of effect size and overlap to simulate how many authors would have found their study to be well-powered. We also examine how many of the observed effects were likely significant based on predicted model overlap. We note that if we use the *observed* effect in a post-hoc analysis, the proportion of studies falling below the MDE is even higher.

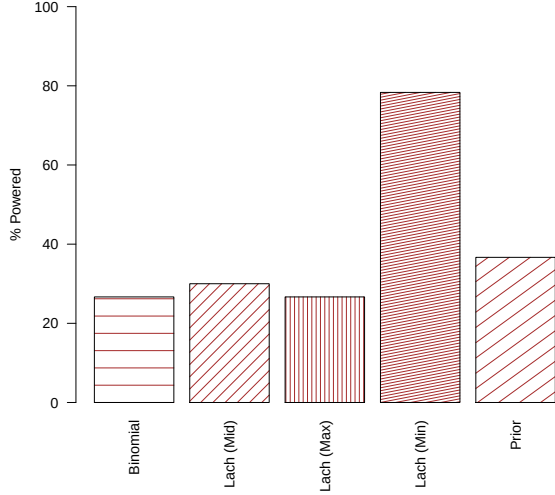


Figure 14: Of the claims of improvement over a given baseline (indicated in text and via bolded values in tables) across 14 papers on the GLUE leaderboard (also seen in Table 10). We find only 26.7% of observed effects met the MDE to the binomial power calculation, 30% met the MDE according to the midpoint calculation of (Lachenbruch, 1992), 26.7% met the MDE when using the upper bound from the (Lachenbruch, 1992) calculation, 78.3% met the MDE when using the most generous (unlikely) assumptions for power according to the MDE (Lachenbruch, 1992) calculation, and 36.7% met the MDE when using the regression-fitted prior of model overlap. Note: this assumes the true population effect *is* the test set effect size. While this is post-hoc power analysis, we felt it may be useful to consider in the context that for a given model comparison on a given test set there is no variance and thus post-hoc power analysis is acceptable. However, for claims that include the entire data distribution this no longer holds and we refer back to the main text.

each class sum to one.

	M1 negative	M1 positive
M2 negative	$p(\text{both neg.})$	$p(\text{only M1 pos.})$
M2 positive	$p(\text{only M2 pos.})$	$p(\text{both pos.})$

Table 11: A contingency table representing the distribution of possible outcomes for two models (M1 and M2) on the instances of a single class of labels. The cells of this table should sum to 1.0 for each class

In addition, we need to assume the true distribution of labels in the data distribution of interest, $p(c)$ for c in $\{1, \dots, C\}$. Given these assumptions, we could then simulate an arbitrary number of datasets from this process. For each instance, we would first sample a true label (c), and then sample the model predictions from the corresponding contingency table. For each simulated dataset, we could then apply the randomization test (using thousands of randomizations). By repeating this process many times, we can directly estimate power for the corresponding assumptions and sample size n .

This process is not particularly efficient, but can still be run relatively quickly on a laptop. The more difficult part is choosing good values for the necessary probabilities. However, such an approach can still be used to test for how sensitive power is to variations in assumptions. It is also possible to make simplifying assumptions, such as that the rate of false positives and false negatives will be the same across classes, or to estimate some parameters from training data, such as the underlying distribution of labels. The same technique can easily be extended to other metrics that depend on the contingency table, such as Matthew’s correlation.

G Additional Details for the BLEU Scores Power Analysis

In this section, we provide further details for the machine translation (MT) data generation procedure as well as an analysis of how power varies for a range of values of P_0 and b_0 , the parameters estimated from the empirical observations.

G.1 Data Generation Procedure

Recall that using the randomization test to determine whether two MT systems are statistically different gives rise to the null distribution of differences in BLEU.¹⁴ If we had access to large amounts

¹⁴The bootstrap is another valid approach to testing for differences between models (Koehn, 2004; Graham et al., 2014; Dror et al., 2018), though note the concerns highlighted

of parallel text, we could instead sample many subsets of real sentences and evaluate the difference between models on those subsets, which allow us to characterize the mean and variance of the difference in model performance. Such estimates could then be used to estimate power directly. Because we do not have access to such data, however, we instead rely on the randomization approach, in which we run several thousand trials where the paired output translations for a subset of the test set samples are swapped. In order to estimate power, we would like to be able to generate many datasets from a data generating procedure, which we can parameterize by various parameters, such as the difference between models. Rather than generating raw text, however, and computing BLEU scores on that, we instead attempt to generate only the data necessary for the randomization test. How can we do this?

In our case, the answer to this question lies in establishing a relationship between individual samples and the permuted set within each trial of the randomization test. This relationship is as follows: *the sum of individual changes to the difference in BLEU, from swapping single samples at a time, closely approximates the net change to the difference in BLEU, from swapping those samples all at once.*¹⁵ Let S be the set of test set samples swapped during a single trial of the randomization test and $R_B(S)$ be the difference in BLEU between the paired outputs after swapping the examples in S . Δ_B is the original difference in BLEU and δ_i is the change to the difference in BLEU from swapping test sample i and leaving all other samples unswapped. Then, we find that,

$$\sum_{i \in S} \delta_i \approx R_B(S) - \Delta_B$$

This relationship is illustrated in Figure 15: Figure 15a shows the difference between two models evaluated on the 2019 test set, and Figure 15b shows the difference between a different pair of models evaluated on the 2018 test set. We found the same relationship is true for the 2017 and 2016 test sets, as well.

Now that we have established a relationship to closely approximate the outcome of each randomization trial, all that remains is to define a distribution from which the individual changes to the

by Riezler and Maxwell (2005).

¹⁵Note that this does not directly solve the problem of computing BLEU at the sentence level (Chen and Cherry, 2014), as it still mimicking the process of evaluating BLEU on a corpus.

difference in BLEU can be sampled. This distribution is a mixture of a Delta distribution at zero and a Laplace distribution. The Delta distribution accounts for the proportion of samples (P_0) such that swapping any of them individually results in no change to the difference in BLEU, i.e. the effect is zero. For the remaining samples, we fit a Laplace distribution, as shown in Figure 16. This Laplace is parametrized by two parameters: location (μ) and scale (b). By fitting this mixture to the individual effects computed from evaluating BLEU differences on many pairs of models, we discover that the variance parameter scales inversely proportional to the size of the dataset. Thus, we report an overall b_0 value for each dataset, such that $b_0 = b_k * n_k$, where b_k is the Laplace scale parameter obtained from dataset k containing n_k samples.

For generating synthetic data, we need to specify μ and b , as well as P_0 . However, because we want the effect of swapping half the non-zero samples from this distribution to equal the difference in BLEU between models, we only use the above fits to estimate b_0 . We thus complete the generative process by assuming values for Δ_B , n , P_0 , b_0 , and setting $\mu = -2 \cdot \Delta_B / (n \cdot (1 - P_0))$ such that the average effect of a random subset of $n/2$ instances is equal to $-\Delta_B$. Table 3 in the main paper shows a range of observed values for P_0 and b_0 .

G.2 Variation in Power Estimates for a Range of Parameter Values

Now that we have defined the data generation procedure, and have estimates for the two parameters, P_0 and b_0 , that are needed to simulate datasets, we can estimate power for a range of values for sample size n and difference in BLEU Δ_B , and see how these estimates vary as P_0 and b_0 change. To provide a concrete example, suppose that we have two machine translation models that we expect will differ by $\Delta_B = 1$ BLEU point. For a dataset of $n = 2,000$ sentences, we assume that the models will perform equally for $P_0 = 0.2$, i.e. 20% of sentences, and will assume a base scale parameter of $b_0 = 26$. To compute power, we would follow the process in Algorithm 1, with the following modifications. On each iteration, we would draw individual changes to the difference in BLEU from the distribution specified above, with $P_0 = 0.2$, $\Delta_B = 1$, $b_0 = 26$, and $n = 2000$. For each such draw, we would apply the randomization test to compute a null distribution, using the sum of in-